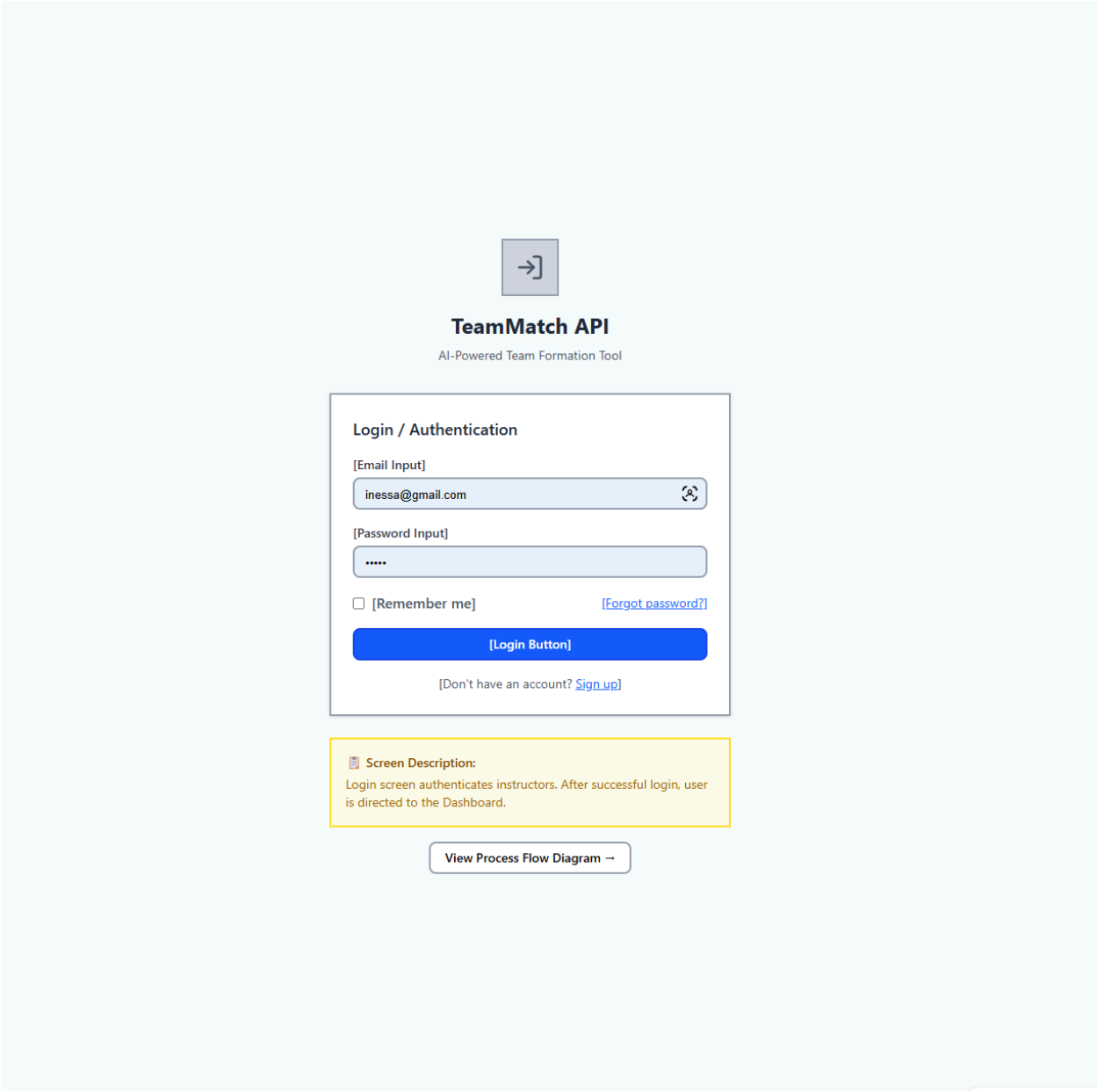
Application design for Teammatch V1

Instructor View:



TeamMatch API Dashboard

[Instructor: Dr. Smith]

[View Flow] [Logout]

3 Total Classes

137 Total Students

29 Teams Created

[Classes & Teams Overview]

[New Team Project]

[Class Name]	[Students]	[Teams]	[Status]	[Actions]
CS 101 - Intro to Programming	45	9	Active	[View Teams] [Edit]
CS 301 - Software Engineering	60	12	Active	[View Teams] [Edit]
CS 401 - Capstone Project	32	8	Draft	[View Teams] [Edit]

[Recent Activity]

- Teams generated for CS 301 - 12 teams created — 2 hours ago
- Student data uploaded for CS 401 - 32 students — 1 day ago
- Teams exported for CS 101 — 3 days ago

[Contribution Tracking & Analytics]

[View Contribution Analytics] [Team Health Dashboard]

[Student Join Team (Demo)] [Student Check-in (Demo)]

Screen Description:

Dashboard shows overview of all classes and existing teams. Users can click "New Team Project" to start the team formation process (→ Upload Data screen) or "View Teams" to see existing teams (→ Team Results screen).

Contribution Analytics

[CS 301 - Software Engineering]

[Back to Dashboard]

42	35	5	2
Total Check-ins (7d)	On Track	Watch	Flagged

[Filters] All Teams | All Statuses | Last 7 days

[Student Contribution Table]

[Student Name]	[Team]	[Last Check-in]	[Hours (7d)]	[Tasks (7d)]	[Evidence]	[Score]	[Status]	[Actions]
Alice Chen	Team Alpha	March 3, 2026	8.5h	3	2 links	92%	On Track	
Bob Smith	Team Alpha	March 3, 2026	7h	2	1 links	85%	On Track	
Carol Davis	Team Alpha	Feb 28, 2026	4h	1	0 links	62%	Watch	
David Lee	Team Alpha	Feb 24, 2026	2h	1	0 links	45%	Flag	
Frank Jones	Team Beta	March 2, 2026	10h	4	3 links	95%	On Track	

API Endpoint:

`GET /instructor/contributions?courseId=cs301`

Returns student contribution data with status flags

Screen Description:

Contribution Analytics shows all students by team with check-in metrics and status badges. Instructors can filter by team, status, or date range. Decision point: "Low contribution detected?" flags students below threshold. Actions include viewing details, messaging students, or requesting updates.

Accessibility:

Table headers clearly labeled, status uses color + text, action buttons have aria labels/titles

Team Health View

[CS 301 - Software Engineering]

80% − + Reset

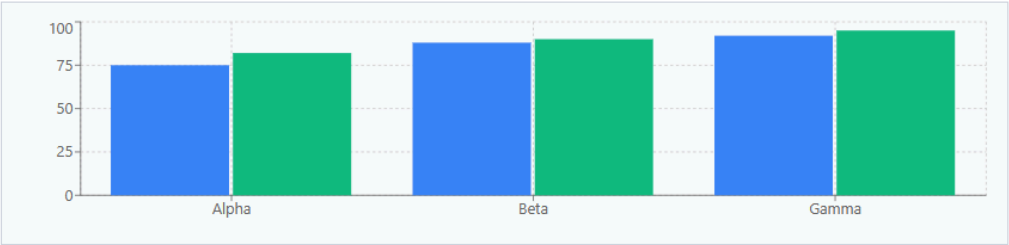
[Select Team] Team Alpha

5
[Team Members]

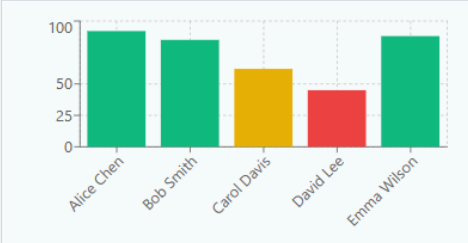
1
[Missing Check-ins (7d)]

74.4%
[Avg Contribution Score]

[Team Balance Comparison]

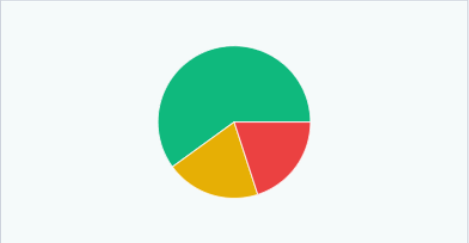


[Contribution Distribution - Team Alpha]



[Individual contribution scores for team members]

[Member Status - Team Alpha]



On Track (3) Watch (1) Flagged (1)

[Team Members - Drill-down]

AC	Alice Chen Last check-in: 2 days ago	92%	[View Details]
BS	Bob Smith Last check-in: 2 days ago	85%	[View Details]
CD	Carol Davis Last check-in: 2 days ago	62%	[View Details]
DL	David Lee Last check-in: 2 days ago	45%	[View Details]
EW	Emma Wilson Last check-in: 2 days ago	88%	[View Details]

[Team Alert]
1 team member(s) have not submitted check-ins in the last 7 days. This may impact team balance and project progress.

Weekly Check-in

[Team Alpha - CS 301]

[View History]

[Weekly Check-in] - Track your contributions to help maintain team balance and accountability.

[Hours Worked This Week] *

8.5

[Tasks Completed] *

Implemented login feature ×

Add a task... +

[What I Worked On] *

Describe your contributions this week...

[Evidence Link] (Optional)

GitHub PR, Google Drive, etc.

Link to GitHub commits, Drive documents, or other work evidence

[Blockers / Challenges]

Any obstacles or challenges you're facing...

[Peer Shoutout] (Optional)

Recognize a teammate's contribution...

[Cancel] [Submit Check-in]

API Endpoint:

`POST /checkins`

Updates contribution score + team balance metrics

Screen Description:

Weekly check-in form captures student contributions. Decision point: "Data Valid?" validates required fields (hours, tasks, description). After submission, contribution scores update and instructor dashboard refreshes. If low contribution detected, system notifies student and flags instructor.

Contribution History

[Team Alpha - CS 301]

[+ New Check-in]

[Your Current Contribution Score]

87% [On Track]

8.8	10	6
Avg Hours/Week	Tasks Completed	Evidence Links

[Score Explanation]

Your score is calculated from hours worked, tasks completed, evidence quality, and peer feedback. Target: 70%+ to maintain good standing.

[Check-in Timeline]

March 3, 2026
8.5h · 3 tasks · 2 links
[Can edit 24h] [Submitted] ▲

[Tasks Completed]

- Implemented login authentication
- Fixed CSS responsive issues
- Reviewed team pull requests

[What I Worked On]

Worked on the authentication system and helped debug CSS issues. Participated in code reviews.

[Evidence Links]

GitHub PR #42: Auth implementation
GitHub commits (week of Feb 24)

[Edit Check-in]

Feb 24, 2026
12h · 5 tasks · 3 links
[Submitted] ▼

Feb 17, 2026
6h · 2 tasks · 1 links
[Submitted] ▼

[Reminder] Your next check-in is due by Friday, March 10, 2026

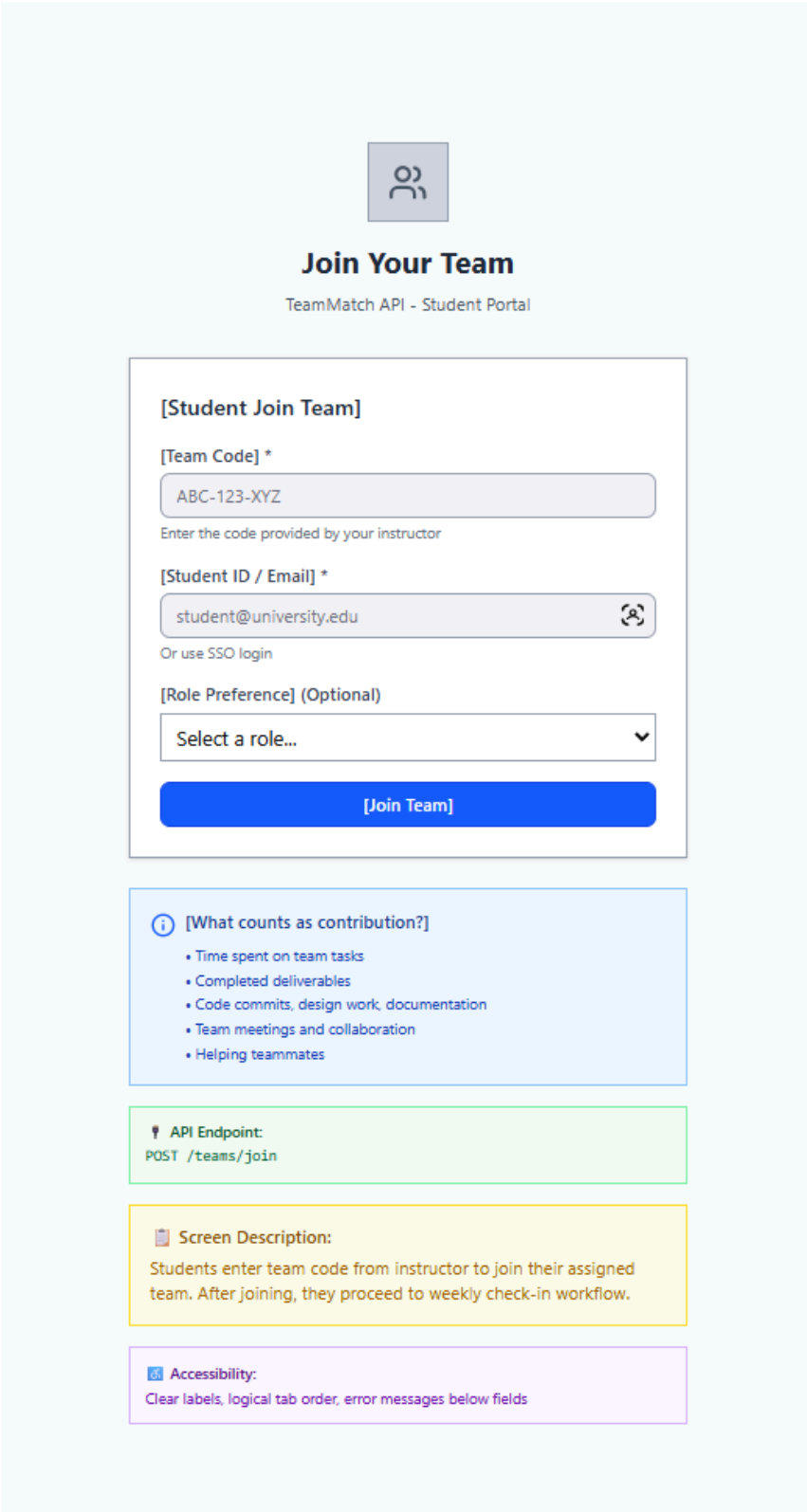
Screen Description:

Contribution History displays student's submission timeline with current contribution score. Students can edit submissions within 24 hours. Score explanation helps students understand their standing.

Accessibility:

Expandable cards use button elements, clear visual hierarchy, status badges with color + text

Student Demo:



Weekly Check-in

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Accessibility:

Clear labels, logical tab order, error messages with icons below fields, ARIA labels on buttons

Contribution History

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[Your Current Contribution Score]

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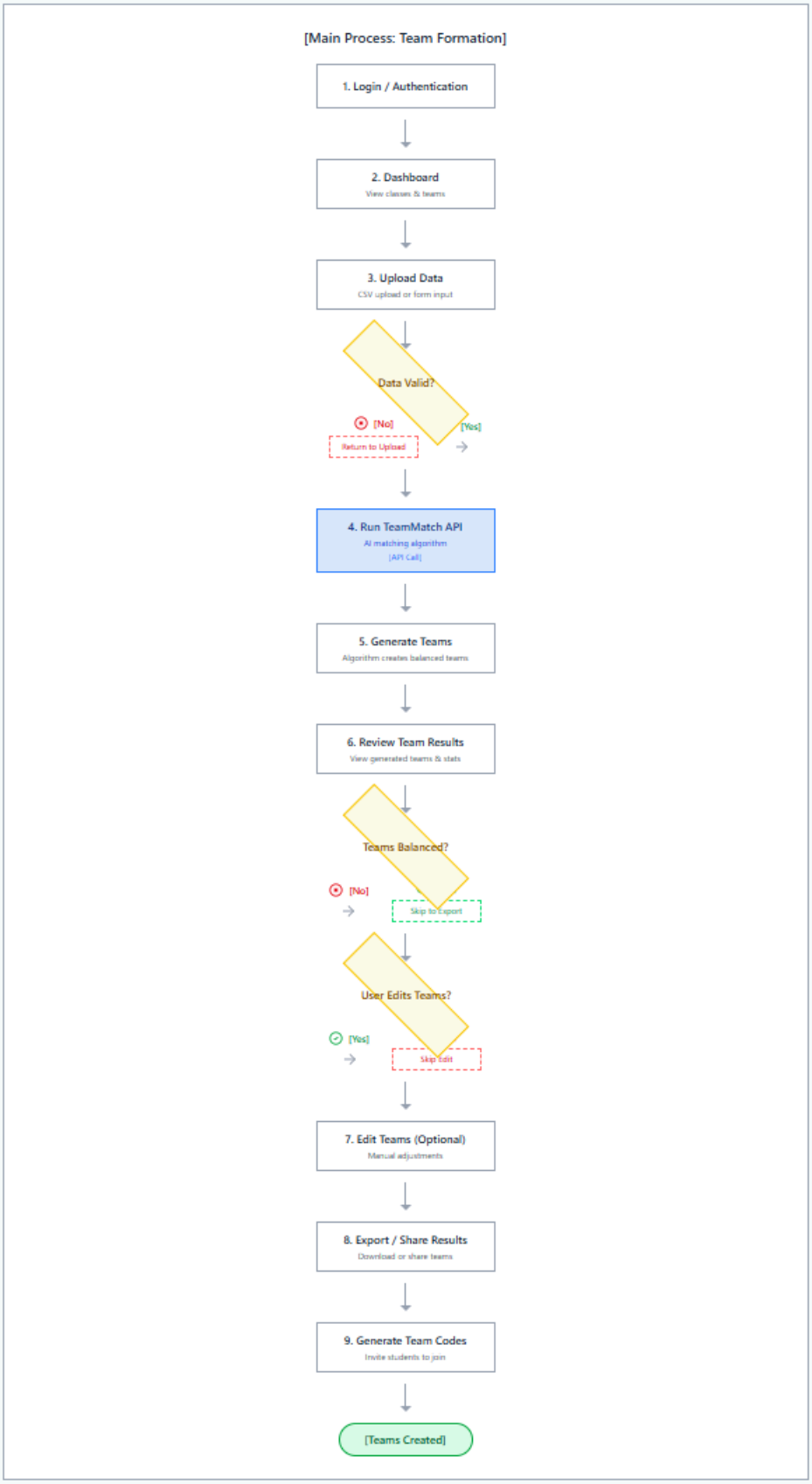
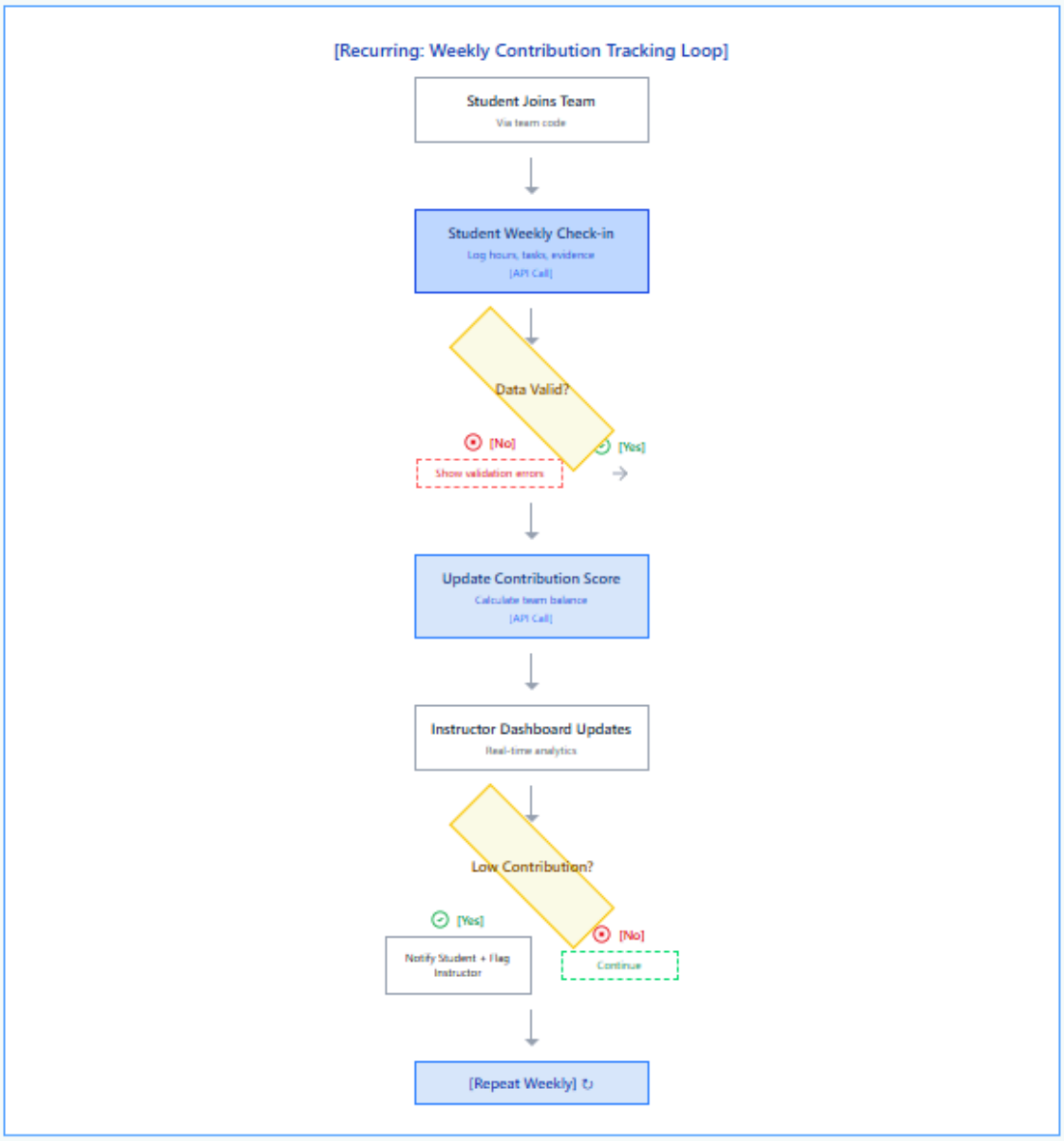
PROCCESS FLOW:

TeamMatch API - Process Flow

Complete user journey from login to team export + ongoing contribution tracking





API Integration Points

Primary APIs:

- POST /api/teams/generate · Runs AI matching algorithm
- POST /teams/join · Student joins team via code
- POST /checkins · Student submits weekly progress
- GET /instructor/contributions?courseId=... · Contribution analytics table
- GET /students/[id]/checkins · Individual student detail view

Optional Future Integrations:

- LMS APIs (Canvas, Blackboard, Moodle) for assignment imports
- GitHub API for commit/PR signals (evidence automation)
- Communication APIs (Email, Slack, Teams)

Decision Points

1. **Data Valid? (CSV/Form)** · System validates CSV format and required fields
2. **Teams Balanced?** · Instructor reviews balance metrics and statistics
3. **User Edits Teams?** · Instructor chooses whether to manually adjust teams
4. **Data Valid? (Check-in)** · Validates hours, tasks, and description fields
5. **Low Contribution Detected?** · Flags students below threshold, notifies student + instructor